

Examples of different types of missing data. Consider the data collected are the worldwide border-crossing data of each country, consisting of the border crossing behavior(quantity, port, measurement, etc.)

1) Missing Completely At Random

Some lost data was due to some unexpected sources of data loss such as DB breakdown, mal-management or even theft. These factors are completely at random.

2) Missing At Random

Some data missing could happen due to reasons like port being remote (such as ports near the north pole in Alaska), or incapability of data collecting such as ports around Syria. In this case, data missing is correlated with another variable.

3) Missing Not At Random

Some countries/ports might not disclose their cross-border behavior due to considerations on national security, and in this case the missing is not at random.

Below are the process of plotting missing data before & after imputation.

The below dataset is a housing price data competition on Kaggle, which contains 79 features and a label(sale price) for Boston housing.

	MSSubClass	MSZoning	LotFrontage	LotArea	Street	Alley	LotShape	LandContour	Utilities	LotConfig	...	PoolArea	PoolQC	Fence	MiscFeature	M
0	60	RL	65.0	8450	Pave	NaN	Reg	Lvl	AllPub	Inside	...	0	NaN	NaN	NaN	
1	20	RL	80.0	9600	Pave	NaN	Reg	Lvl	AllPub	FR2	...	0	NaN	NaN	NaN	
2	60	RL	68.0	11250	Pave	NaN	IR1	Lvl	AllPub	Inside	...	0	NaN	NaN	NaN	
3	70	RL	60.0	9550	Pave	NaN	IR1	Lvl	AllPub	Corner	...	0	NaN	NaN	NaN	
4	60	RL	84.0	14260	Pave	NaN	IR1	Lvl	AllPub	FR2	...	0	NaN	NaN	NaN	
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	
1455	60	RL	62.0	7917	Pave	NaN	Reg	Lvl	AllPub	Inside	...	0	NaN	NaN	NaN	
1456	20	RL	85.0	13175	Pave	NaN	Reg	Lvl	AllPub	Inside	...	0	NaN	MnPrv	NaN	
1457	70	RL	66.0	9042	Pave	NaN	Reg	Lvl	AllPub	Inside	...	0	NaN	GdPrv	Shed	
1458	20	RL	68.0	9717	Pave	NaN	Reg	Lvl	AllPub	Inside	...	0	NaN	NaN	NaN	
1459	20	RL	75.0	9937	Pave	NaN	Reg	Lvl	AllPub	Inside	...	0	NaN	NaN	NaN	

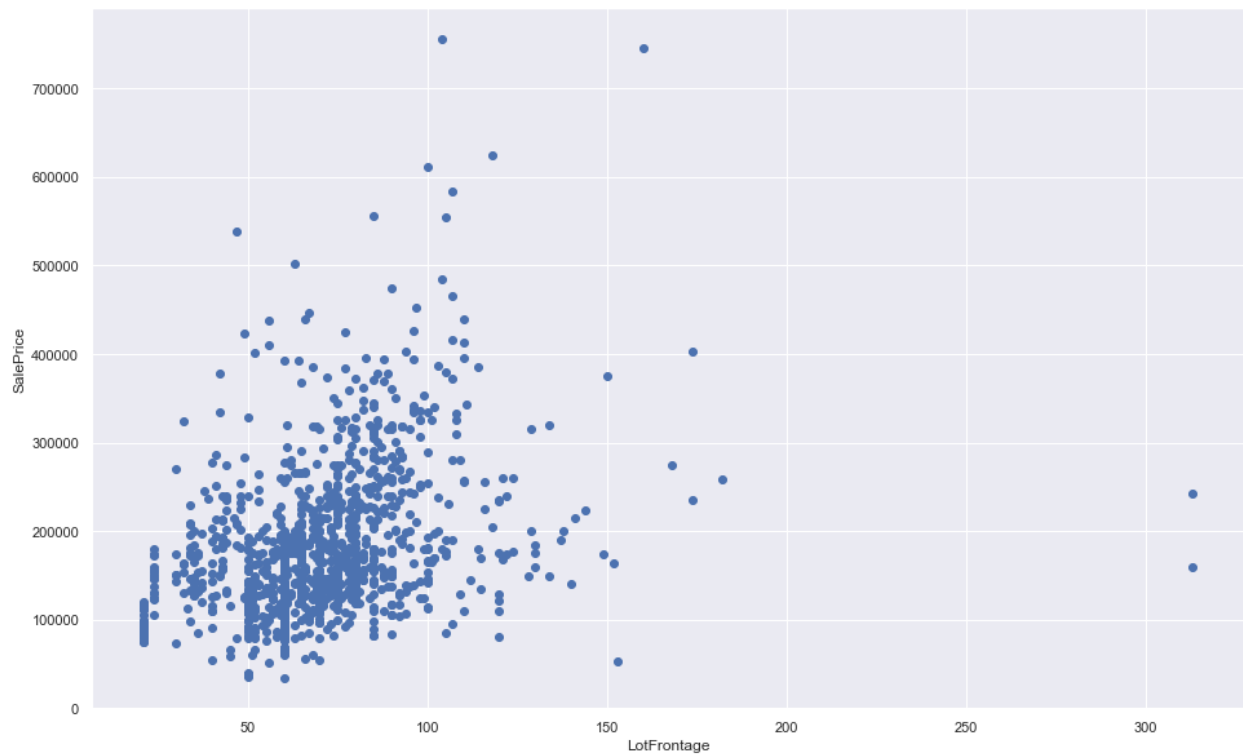
1460 rows x 80 columns

Columns that contain missing values:

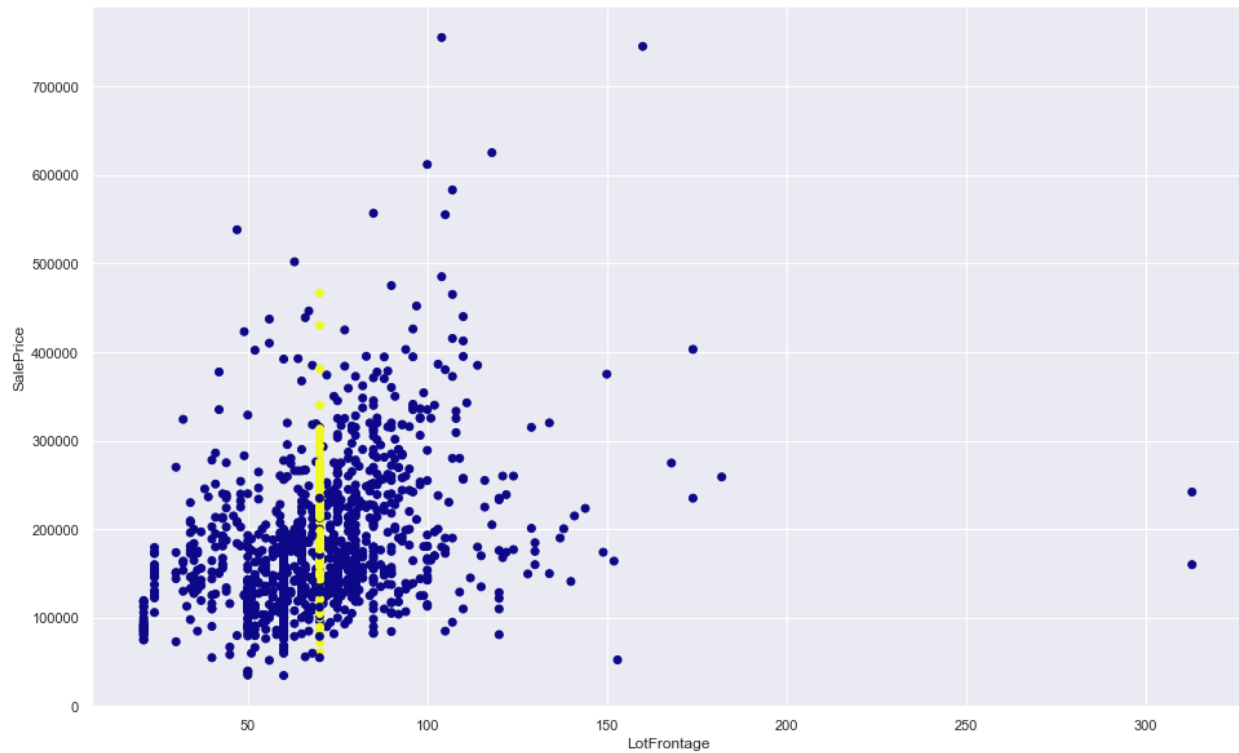
	cnt
LotFrontage	259
Alley	1369
MasVnrType	8
MasVnrArea	8
BsmtQual	37
BsmtCond	37
BsmtExposure	38
BsmtFinType1	37
BsmtFinType2	38
Electrical	1
FireplaceQu	690
GarageType	81
GarageYrBlt	81
GarageFinish	81
GarageQual	81
GarageCond	81
PoolQC	1453
Fence	1179
MiscFeature	1406

Try to impute missing values for numeric column “LotFrontage” and categorical column “FireplaceQu”.

For feature “LotFrontage”, below is a plot between “LotFrontage” and the label.

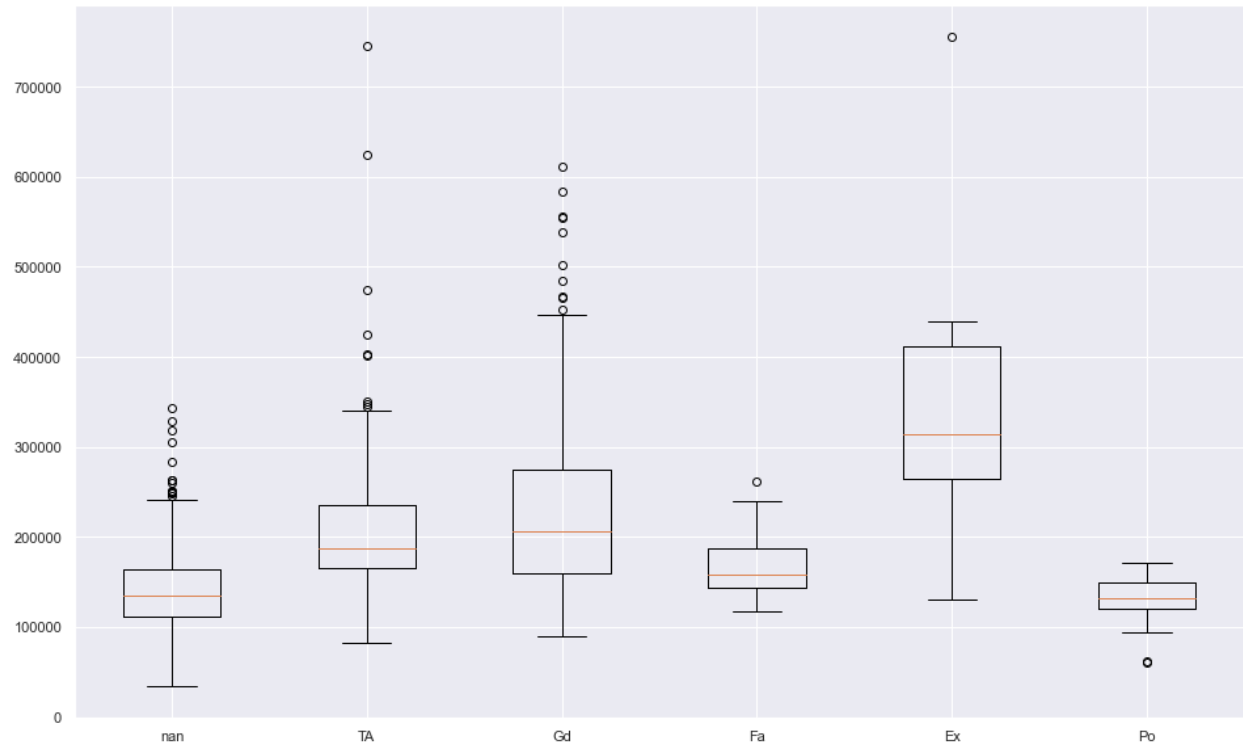


After imputing missing values with LotFrontage mean, the scatter plot looks like below.

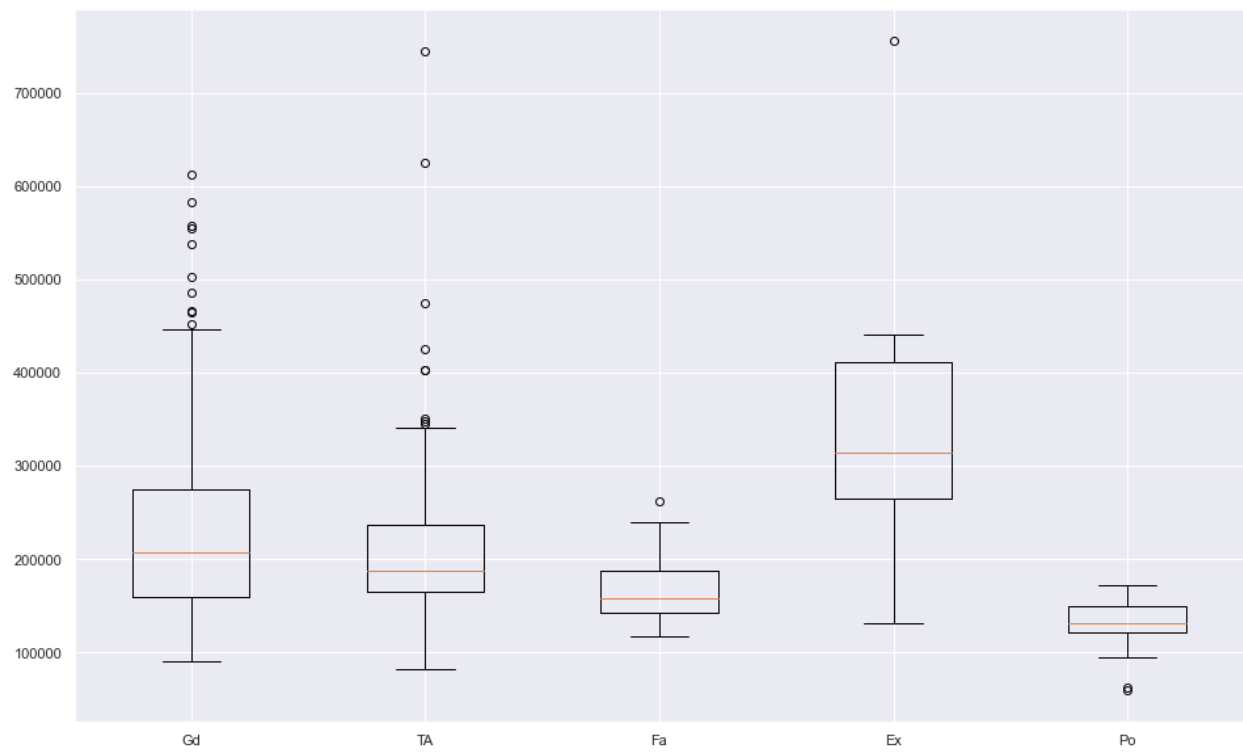


The yellow dots represent original missing values which are imputed with the mean.

For categorical column “FireplaceQu”, the original box plots of different categories are shown as below.



After imputing missing values with the FireplaceQu mode, the box plots are updated below.



As shown above, the “nan” category disappears, which contributes to value “Gd” (though the difference in boxplot is not obvious enough to distinguish).